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Using Deep Learning for Spatial and Temporal Analysis of Wildfire Start and Progression

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***Abstract*— Wildfires pose a significant threat to ecosystems, public safety, and property, and understanding the spatial and temporal patterns of their occurrence is of paramount importance for mitigating their devastating impacts. In this study, we address the critical task of wildfire start and progress prediction by leveraging cutting-edge deep learning techniques. We obtain data from remote-sensing and ground based input sources to analyze patterns for next 24 hours using past 24 hours of data. Real time wildfire prediction requires a multi dimensional model, that is able to learn and retain information from our data set. Our approach employs a novel deep learning architecture, which captures both spatial and temporal dependencies in wildfire data, grants us results that provide substantial breakthroughs in using ML to predict wildfire start and progress.**

I. Introduction

The global landscape is witnessing a profound transformation driven by the shift in global temperatures, with far-reaching consequences on our environment and public well-being. Notably, this transformation has led to an alarming surge in wildfires, engendering a chain of adverse effects reminiscent of the air pollution crisis. As populations gravitate towards minimizes carbon emissions, the lingering effects of years of pollution continue to alter the physical landscape through natural wildfires.

This wildfire proliferation, has led to over 2,400 deaths in the past three decades. In recent years, wildfires have become increasingly prevalent, exacerbating environmental degradation and threatening biodiversity. As highlighted 2020 California wildfire season, approximately 9,900 wildfires occurred ranging in size and acres burned — putting the standing infrastructure to the test. [Keeley and Syphard, 2021]

The consequences of wildfires extend far beyond the immediate and visible destruction of landscapes. Much like wildfires's insidious ability to infiltrate human respiratory tracts, wildfires emit particulate matter and pollutants that can lead to premature mortality, particularly among individuals with pre-existing heart and lung conditions. This peril transcends age groups, adversely affecting the health of communities, and exacerbating conditions such as Chronic Obstructive Pulmonary Disease (COPD), asthma, pulmonary fibrosis, and pneumonia. [Muthukumar et al., 2021b]

In essence, the surge in wildfires underscores the imperative for concerted efforts to mitigate the detrimental effects on human health, ecosystems, and habitat preservation. Forming potential plans for evacuation and safety in advance of a wildfire is integral to preventing the loss of human life, which can be achieved through mechanisms that inform us of wildfire risk and progression beforehand. [Jaiswal et al., 2022]

Our investigation into wildfire dynamics empowers us to gain comprehensive insights into this global challenge, enabling us to anticipate future patterns and formulate proactive solutions. Our research employs an advanced deep learning architecture that combines spatial and temporal modeling, allowing us to predict both wildfire starts and progression.

We utilize multi source data in our deep learning approach, which forms the basis for a comprehensive predictive model. We leverage g atmospheric remote sensing data, and high-resolution satellite imagery capturing wildfire and meteorological features. The heart of our complex deep learning framework uncovers various features across our data and stores the key components for the prediction formulation.

II. Methodology

For our wildfire start and progression prediction of Los Angeles, we rely on various remote sensing instruments and data sets sourced by NASA and the California Energy Comission. We utilize remote-sensing wildfire indicators, power line outlines, ground-based soil data, and ground-based meteorological gridded data in our deep learning architecture.

We use a Los Angeles County bounding box of 40 km by 40 km area, between the latitudes of $33.5°N - 34.5°N$ and longitudes of $117.5°W - 118.75°W$. [Muthukumar et al., 2021a]

To study our Los Angeles County defined area, we analyze various factors correlated with the climate and topographical features of the region. We primarily control for the presence of elevation in regions of northern LA county, granted its ability to influence the dispersion of a wildfire over a specified area of vegetation/forest. One of the longest active fires during the 2020 wildfire season across California was the Ranch 2 fire in LA County which lasted for 53 days, and burned over 4,237 acres. Wildfires in Los Angeles are fueled by the Santa Ana winds [Westerling et al., 2004]which are seasonally found in the fall and winter due to excessive levels of humidity and pressure from the coast. In tandem with these winds is rainfall

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which promotes unusual vegetation growth and increases the amount of potential area at a fire risk.

By combining direct wildfires historical data and indirect data-characterizing factors that are associated with wildfires, our remote-sensing and ground-based data forms a comprehensive approach to the hourly prediction of wildfires. Air pollution cannot be effectively linearly predicted using a statistical approach because of the way it evolves and changes from various environmental factors [Rosenlund et al., 2008]. Instead, we provide the Deep Learning architecture with data to deduce how the environment and topographical features affect patterns of wildfires spatiotemporally in Los Angeles county. The various remote-sensing instruments are recorded at differing sea levels, urban structures, and topographies, thus providing our model with robust multifaceted information.

A. Dataset

We scaled our remote sensing and ground-based input data sources to a 40px x 40px resolution, as representation image of Los Angeles county. We utilized three years of hourly data from January 1, 2018 to December 31, 2020. We collected 26304 samples corresponding to 24 hourly samples for the 1096 days of data from January 1, 2018 to December 31, 2020.

To account for the possible ways a wildfire can move across a region, we included data from the NASA Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) satellite [Abrams et al., 2015]. ASTER informs of various elevation or extrusion across the area, and allows us to capture wildfire paths across the terrain. We can also note that if a wildfire is moving downhill it will move at a much faster pace compared to if it is moving uphill, improving the accuracy of our time-based predictions.

Fig. 1: Ground Truth Label for our Deep Learning Model: NASA Multiangle Imaging Spectrometer

Identifying the prevalence of wildfire risk factors is crucial to our predictive model, and to accomplish this we rely on data from the California Energy Commission. (CEC) The CEC provides us with a map that demarcates power transmission lines that run across LA County, and incorporate details about their directional path and length. Power transmission lines can potentially serve as the cause of a wildfire during catastrophic natural events, making it a leading concern to consider when evaluating where a wildfire can begin. [Bandara et al., 2023]

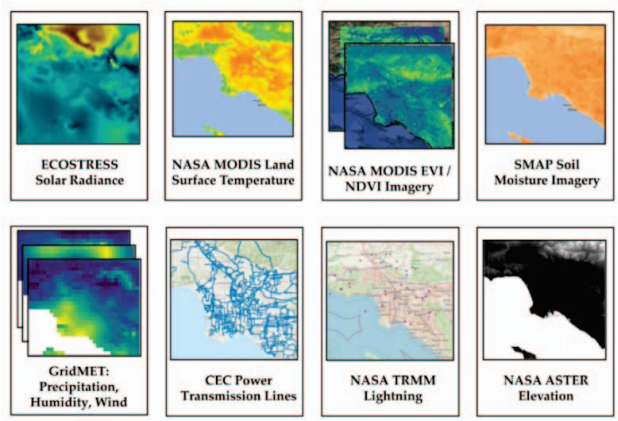

Fig. 2: Various input features used in our Deep Learning predictive model

Following a similar line of reasoning to the CEC Power Transmission Lines, we have incorporated data from the NASA Tropical Rainfall Measuring Mission. (TRMM) The mission satellite carries multiple different monitors that observe information about weather patterns across the globe, and we utilized the Lighting feature of the satellite in our model. Observing lightning patterns across the Los Angeles County region is informative because the occurrence of high intensity directly onto vegetation can lead to a wildfire outbreak, causing severe damage.

The NASA MODIS Enhanced Vegetation Index and Normalized Difference Vegetation Index are a part of a remote imaging spectrometer and are derived using atmospherically-corrected reflection. The instrument uses light reflection to measure the presence of vegetation on the land, and identifies regional boundaries where wildfire outbreak risk is high. The product typically has varied resolutions, but are high enough to help our model make conclusive predictions. We also use the Land Surface Temperature band of the MODIS satellite, which uses remote-sensing information to geographically map regions of LA county that are warm compared to the general average. We observe a direct correlation between temperature and wildfire risk, making the land surface temperature results particularly useful.

We incorporated ground-based meteorological data into our model, drawing from the gridMET meteorological dataset. gridMET employs climatically-aided interpolation to map climate variables onto a geographic grid [Abatzoglou, 2013]. Among the key variables from gridMET that we leveraged for our predictive model, one stands out: precipitation accumulation. This variable offers insights into local climate aridity and helps us establish a link between wildfires and precipitation patterns.

In addition to precipitation, we harnessed wind-related data from gridMET, particularly wind speed and wind direction.

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Wind speed information is invaluable for comprehending the dynamics of wildfire propagation, shedding light on how fires spread in our specific geographic region. Equally critical is the wind direction data, which provides a comprehensive view of the directional movement of future wildfire patterns.

Through the Ecosystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS) we were able to include a band for solar radiance, which serves as the measure of radiance emitted by the sun and diffused on a land region. In other words it measures the amount of sunlight hitting our region of study, and with that information we can determine if the direct sunlight can be a cause for concern.

Finally, we included the Soil Moisture measurement from the NASA Soil Moisture Active Passive (SMAP) instrument. The remote sensing instrument pushes radio waves into parts of the Earth, and detects if the ground emits a certain level of radio waves or echo back. It provides a spatial resolution of 3km, and it helpful in understanding if the region is dry and at risk of burning. [Krueger et al., 2022]

III. Results

Our deep learning approach can be broken up into two core concepts: wildfire start risk prediction & wildfire progress prediction. For our wildfire start risk prediction, we utilized remote sensing satellite data in combination with ground-based geographical data; for our wildfire progress prediction we informed our model that a wildfire had occurred in a specific region and utilized similar features to predict the path of the wildfire.

Our cutting-edge model forecasts hourly wildfire probabilities across LA County, quantifying the risk as a percentage per square kilometer. It leverages a novel approach, analyzing 24 historical frames to predict the likelihood of wildfires in the next 24 hours. This prediction is fueled by a fusion of multisource satellite imagery, meteorological data, and remote-sensing information on previous wildfire occurrences. [Muthukumar et al., 2022]

To quantify how our wildfire start risk prediction did against actual NASA Multiangle Image Spectrometer (MISR) fire data, we utilize precision and recall scores. Precision and recall help us identify how many correct predictions were made from the model, and of the correct predictions how many of them were fully correct. This method of validation typically appears useful in the case of a classification task, and is suitable for identifying if a wildfire occurred at a specific site.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives + False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives + False Negatives}}$$

We also used the Root Mean Square Error (RMSE) and Normalized Root Mean Square Error (NRMSE) error. RMSE and NRMSE are calculated as

$$\text{RMSE} = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}; \ \text{NRMSE} = \frac{\text{RMSE}}{\bar{y}}$$

where n is the number of observations, $\hat{y}$ is the predicted value, y is the ground truth, and $\bar{y}$ is the mean of the test data.

We tuned our model's parameters and weights to fit the data set, and utilized Adam, an iteration of Gradient Descent, to do so.

A. Wildfire Start and Progress Result Frames

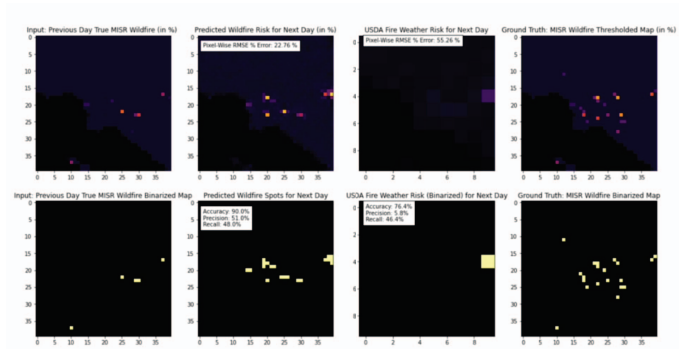

Fig. 3: Deep Learning Model Results for Wildfire start risk prediction

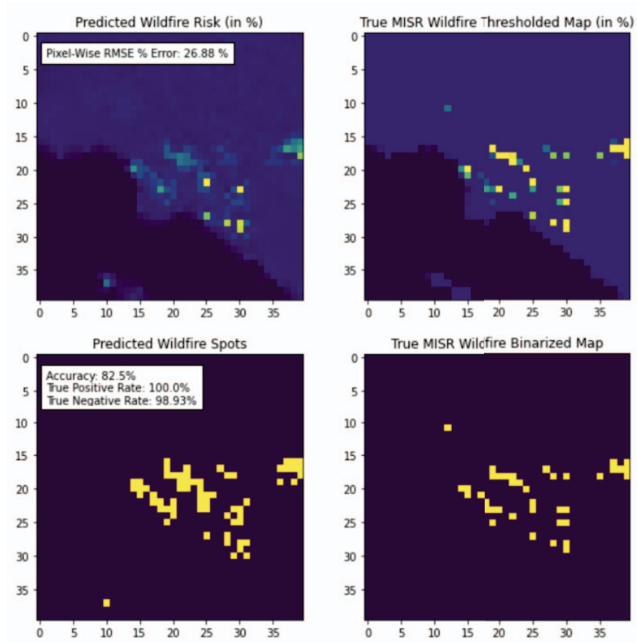

Fig. 4: Deep Learning Model Results for Wildfire Progress Prediction

In Figure 3 are our results from the deep learning model which demonstrate the accuracy of the prediction, directly

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compared with the actual wildfire locations. The yellow markings on each grid are representative of wildfire start locations. We notice that the model performance at an accuracy of 90%, RMSE Error of 22.76%, Precision of 51%, and a Recall of 48%.

In Figure 4, the image of the top left is representative of our model's wildfire progress predictions, top right is our MISR ground truth label, bottom left is our model's binary predictions for if a wildfire occurred, and bottom left is the binary MISR ground truth label

For wildfire progress model, we had a pixel-wise RMSE score of 26.88 % in the given frame. For the binary mapping of wildfires, we had an accuracy of 82.5%, True Positive Rate of 100%, and a False Negative Rate of 98.93%.

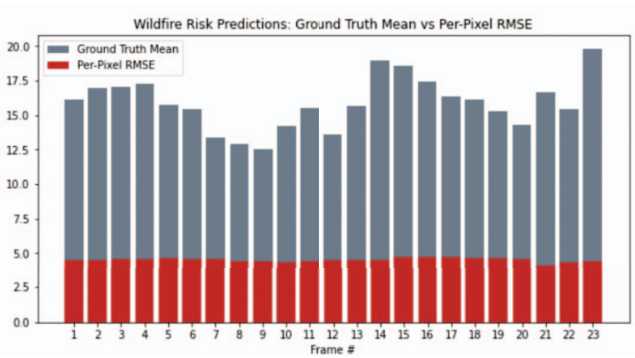

Fig. 5: Per-Pixel RMSE (red) graphed with Ground Truth Mean RMSE (gray) for Frames 1-23

IV. Conclusion

Through our spatiotemporal study, we adeptly utilized modern deep learning methodologies to generate accurate hourly wildfire predictions in the defined region of Los Angeles County. Our comprehensive approach drew from an array of data sources, encompassing ground-based meteorological data, and intelligence derived from remote-sensing satellite instruments. [Muthukumar et al., 2020a]

Our deep learning model is retentive and able to predict wildfire patterns sequentially over a defined region of land. Throughout our model architecture, we carefully considered how each data source could be applied including remote sensing measurements involving solar radiance, soil moisture, elevation, meteorological factors, lightning, land temperature, vegetation and ground-based mappings for power transmission lines.

We validated our results with three different metrics: Per-Pixel RMSE, Precision, and Recall. By validating our results we found that we were gradually able to reduce the per-pixel RMSE score frame by frame. We performed a comparison with two different labels, the ground truth MISR wildfire data, and USDA Fire Weather Index data. Using two different comparison sources gave us a clear evaluation of the model performance, and direction for future research in refining model results.

Additionally, we note that our model is able to fit two different purposes and re-train to provide accurate predictions both for wildfire start and progress. It sets results in a completely unique domain: using deep-learning techniques to predict wildfires. Our advanced structure can be used alert Los Angeles county researchers in the field about potential wildfire risks, and aid the evacuation process for any period of time.

Fig. 6: Precision and Recall Scores for our Deep Learning Approach when compared with the USDA Fire Weather Index

V. Future Work

In future, we can expand our research to more comprehensively study common wildfire site locations in the Los Angeles County area. We can dive deeper in the issues causing repeated wildfires, and increase the size of our data set to accommodate for more wildfire indicator variables.

Wildfires can have impacts that lead on to several different respiratory and life-threatening diseases. We can use the data we have on wildfires to further understand how its dispersion causes air pollution, and the increase in disease risk for individuals with premature conditions. Aside from meteorological factors, we can also analyze how forest management and preservation services can help try to mitigate a wildfire occurrence.

From a deep learning perspective, we can aim to increase the resolution of model by using inputs that provide a higher spatial resolution. We would potentially be able to capture more details about the LA County topography through a more close up study of the defined region. We currently operate at a 40px x 40px resolution, and we hope to move into further research that delivers a higher resolution. [Muthukumar et al., 2020b] We could also expand the study to predict results at a more frequent basis, instead of every hour to leverage more accurate results.

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